

PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Technical Architecture

The solution is implemented on the ServiceNow platform. The administrator interacts with ServiceNow through its web interface. The Excel file is supplied as the source, ServiceNow loads it into an Import Set staging table, and a Transform Map moves the data into the custom Employee Test table. Coalesce supports insert/update decisions, while ServiceNow reports and the dashboard provide the presentation layer for the resulting employee information.

Logical Architecture – Employee Data Import and Analytics

Infrastructure demarcation: the source spreadsheet is an external/local file supplied to the ServiceNow instance; the import processing, target data, reports and dashboard are handled within ServiceNow. No separate external API or machine-learning component is used in the implemented project.

Table-1: Components & Technologies

Table-2: Application Characteristics

Technology Selection Rationale

ServiceNow Import Sets provide a controlled staging point for spreadsheet data before it reaches the target employee table.

Transform Maps provide reusable source-to-target field mappings and make the import process repeatable.

Coalesce addresses the project's main data-quality concern by preventing unnecessary duplicate employee records.

Native Reports and Dashboards provide a straightforward way to monitor the resulting employee information without introducing another reporting platform.

The solution does not require external APIs or machine-learning services for the implemented scope.

References

The architecture is documented according to the structure of the supplied Technology Stack template. The implementation-specific technology choices above are based on the completed ServiceNow project documentation.

Date	03 October 2026
Team ID	SWTID-2026-8740
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Source	Process	Data Store	Analytics	User
Excel Spreadsheet	ServiceNow Load Data	Employee Import (u_employee_import)	Reports	HR / ServiceNow Administrator
Employee rows	Transform Map	Employee Test (u_employee_test)	Employee Analytics Dashboard	Web UI
	Coalesce	Insert / Update / Ignore	Department • Location • List	
External/local file	ServiceNow platform	ServiceNow managed data	ServiceNow reporting	Authorized user

S.No	Component	Description	Technology
1	User Interface	Web interface used by administrators to load data, configure transformations, view records and access reports/dashboard.	ServiceNow Web UI
2	Application Logic-1	Loads the spreadsheet into a staging/import table.	ServiceNow Load Data / Import Set
3	Application Logic-2	Maps source fields to target fields and executes the data transformation.	ServiceNow Transform Map
4	Application Logic-3	Identifies matching employee records during repeated imports and supports reporting/visualization.	ServiceNow Coalesce, Reports and Dashboard

S.No	Component	Description	Technology
5	Database	Stores the staging and target employee records used by the application.	ServiceNow platform tables: u_employee_import and u_employee_test
6	Cloud Database	Managed data persistence for the ServiceNow application.	ServiceNow-managed platform database
7	File Storage	Source file used for employee-data import.	Excel (.xlsx) file supplied by the administrator
8	External API-1	No external API is required for the implemented import workflow.	Not applicable
9	External API-2	No external API is required for the implemented import workflow.	Not applicable
10	Machine Learning Model	No machine-learning model is part of the implemented solution.	Not applicable
11	Infrastructure (Server / Cloud)	Application execution and data processing are provided through the ServiceNow platform.	ServiceNow cloud/SaaS platform

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is used. The project uses ServiceNow's native configuration features.	ServiceNow platform
2	Security Implementations	Access is controlled through ServiceNow's roles/ACL mechanisms. The project documentation specifically uses the pa_dashboards ACL and Admin Overrides setting for dashboard creation.	ServiceNow ACL / role-based access
3	Scalable Architecture	The design separates source staging from the target employee table and uses a reusable Transform Map. This allows repeated imports without redesigning the basic workflow.	Import Set + Transform Map + target table
4	Availability	The solution is hosted on the ServiceNow platform rather than a locally deployed application. Availability therefore depends on the configured ServiceNow instance.	ServiceNow cloud platform
5	Performance	Bulk spreadsheet loading and transformation reduce the need for manual record-by-record entry. No project-specific requests-per-second or cache benchmark was defined.	ServiceNow Import Set / Transform Map